

Counting processes

Covered by chapter 4.1 in the textbook and these notes.

In this lecture we will learn how to simulate certain continuous time stochastic processes.

We focus on simulating the following “counting” processes:

- Poisson process
- Renewal process
- Nonhomogeneous Poisson process

Such processes are useful as “input” when simulating models that evolve over time.

Stochastic processes

Stochastic processes are “systems” which evolve randomly, often as a function of time. In mathematical terms, we study some random variable X which evolve in “time”.

Continuous time processes:

If the random variable is updated continuously we call the process a *continuous-time* process and write the process as $\{X(t) : t \geq 0\}$.

Examples

$X(t)$ = the production volume from an oil well at time t .

$Y(t)$ = the number of customers waiting to be served in a bank at time t .

$W(t)$ = the number of fire detectors in a building functioning at time t .

$N(t)$ = the number of events in a process until time t .

Discrete time processes:

If the random variable is updated at discrete steps we call the process a *discrete-time* process and write the process as $\{X_n : n \in I\}$ where I is a countable index set, for instance the natural numbers such that $\{X_n : n = 0, 1, 2, \dots\}$

Examples

X_n = the position of your favourite football team after round n .

Y_n = indicator of whether it is rain or not on day n .

W_n = the number of bugs in program code version n .

State space

The set of possible values of X in a stochastic process is called the *state space*. A state space can either be discrete or continuous.

Counting process:

Let $N(t)$ = the number of events in $[0, t]$.

$\{N(t), t \geq 0\}$ is then called a *counting process*.

Notice in particular that “the number of events which occur in an interval $[a, b]$ ” can simply be written: $N(b) - N(a)$.

Notice; discrete state space $[0, 1, 2, \dots)$.

We shall now learn about some common counting process models, including how to simulate data from such models. Such processes have numerous applications in areas like:

Reliability modelling, queue modelling, network models, computer games, traffic models, physics, etc.

Homogeneous Poisson process

A counting process is a homogeneous Poisson process with *intensity* λ if:

1. $N(0) = 0$
2. The numbers of events that occur in disjoint intervals are independent. I.e. $N(b) - N(a)$ indep. of $N(d) - N(c)$ if $[a, b]$ and $[c, d]$ are disjoint. (Independent increments).
3. $P(N(s + \Delta s) - N(s) = 1) \approx \lambda \Delta s$.
(Stationary increments).
4. $P(N(s + \Delta s) - N(s) > 1) \approx 0$.

It can be shown that $X = N(s + t) - N(s)$ has a Poisson distribution with expectation λt . I.e.:

$$f(x) = \frac{(\lambda t)^x}{x!} e^{-\lambda t}, \quad x = 0, 1, 2, \dots$$

and $\mu = E(X) = \lambda t$ and $\text{Var}(X) = \lambda t$.

Notice that λ represents the expected number of events per unit i.e. the *intensity* of the process.

Properties

Let T_1 be the time until the first event in a Poisson process. We then have that:

$$\begin{aligned} P(T_1 > t) &= P(\text{no events in } [0, t]) \\ &= \frac{(\lambda t)^0}{0!} e^{-\lambda t} = e^{-\lambda t} \end{aligned}$$

I.e. $F(t) = P(T_1 \leq t) = 1 - e^{-\lambda t}$, and then the probability density of T_1 becomes $f(t) = F'(t) = \lambda e^{-\lambda t}$ which is the exponential distribution.

I.e. the time until the first event in a Poisson-process has an exponential distribution. Similarly it can be shown that all the times *between* events in a Poisson process $T_1, T_2, T_3, \dots$ has an exponential distribution and are independent (e.g. the time between each new visitors to a web-page, the time between each time a fire brigade is called out, etc.)

The times between events in a counting process, $T_1, T_2, T_3, \dots$, are often called *inter-arrival times*.

Further properties

Homogeneous Poisson processes have several other properties, we shall briefly mention two:

- The time until event number k occurs in a Poisson process, $S_k = T_1 + T_2 + \dots + T_k$, has a gamma distribution with parameters $\alpha = k$ and $\beta = 1/\lambda$.
- If we know the number of events n in an interval of length t , the event times are uniformly distributed over the interval. This is useful for simulations.

Simulating homogeneous Poisson processes

It is easy to simulate an HPP with intensity λ , and it can be done in several ways. Two possibilities are:

- Simulate interarrival times $T_1, T_2, \dots$ from the exponential distribution with parameter λ .
- First simulate the number of events, n , in an interval $[0, t]$ from the Poisson distribution with expectation λt . Then simulate the arrival times inside this interval by simulating n numbers from the Uniform $[0, t]$ distribution and sort the numbers.

See R-code: `stochastic_process_examples.R`

Renewal process

A renewal process (RP) is an easy extension of the HPP where the interarrival times, $T_1, T_2, T_3, \dots$, are iid from any continuous distribution.

I.e., the HPP is a special case of the RP with iid exponentially distributed interarrival times.

Simulating renewal processes

We can easily simulate an RP by simulating iid interarrival times from the distribution for the interarrival times.

See R-code: `stochastic_process_examples.R`

Nonhomogeneous Poisson process

A counting process is a nonhomogeneous Poisson process with *intensity* $\lambda(t)$ if:

1. $N(0) = 0$
2. The numbers of events that occur in disjoint intervals are independent. I.e. $N(b) - N(a)$ indep. of $N(d) - N(c)$ if $[a, b]$ and $[c, d]$ are disjoint. (Independent increments).
3. $P(N(s + \Delta s) - N(s) = 1) \approx \lambda(s)\Delta s$.
4. $P(N(s + \Delta s) - N(s) > 1) \approx 0$.

It can be shown that $N(s+t) - N(s)$ has a Poisson distribution with expectation $\int_s^{s+t} \lambda(u)du$.

I.e. the difference from the HPP is that the NHPP allows the intensity to vary over time.

Simulating nonhomogeneous Poisson processes by thinning

There are several ways to simulate nonhomogeneous Poisson processes. One approach is using a “thinning” method similar to the acceptance-rejection method. This method applies when $\lambda(t)$ is bounded on the interval of interest.

Let $[a, b]$ be the interval of interest. If

$$\lambda(t) \leq \lambda_{max} \text{ for all } t \in [a, b]$$

then we can:

1. Simulate an HPP with intensity λ_{max} , giving the simulated arrival times $s_1, s_2, \dots, s_N$.
2. Simulate N variables $u_1, \dots, u_N$ from the $U[0, 1]$ distribution.
3. For each simulated arrival time s_i , accept the time with probability $\lambda(s_i)/\lambda_{max}$, i.e. accept the time if $u_i \leq \lambda(s_i)/\lambda_{max}$, otherwise delete this time.
4. Return the remaining arrival times.

The proof for the thinning algorithm is similar as the proof for the acceptance-rejection method.

Example

The number of cars arriving at a roundabout during a two hour interval in the morning is described by a nonhomogeneous Poisson process with intensity $\lambda(t) = 200 + 190 \sin(2t - 1)$. Plot the intensity function and simulate realizations of the process.

See R-code: `stochastic_process_examples.R`

Example:

The number of visitor to a news page on the web is modelled as a nonhomogeneous Poisson process with intensity $\lambda(t) = ae^{-bt}$.

Let the time scale be hours and consider the case with $a = 800$ and $b = 0.1$.

Calculate both exactly and by simulations:

- The expected number of visitors during the first 10 hours.
- The probability of more than 5000 visitor during the first 10 hours.

Recall $\lambda(t) = ae^{-bt}$ and considering the time interval $[0,10]$.

Exact calculations:

Let X =Number of visitors in first 10 hours.

Then $X \sim \text{Poisson}(\int_0^{10} \lambda(t)dt)$.

Further:

$$\int_0^{10} \lambda(t) dt = \int_0^{10} a \exp(-bt) dt =$$

$$= \frac{a}{b} - \frac{a \exp(-10b)}{b} \approx 5056.964469$$

Now for the sought probability;

$$\begin{aligned} P(X > 5000) &= 1 - P(X \leq 5000) = \\ 1.0 - \text{ppois}(q=5000, \text{lambda}=5056.964469) &= \\ = 0.786156. \end{aligned}$$

Alternatively, a normal approximation would also be OK.

See R code: `stochastic_process_examples.R`

Random walk

The simplest form of a random walk is a discrete time stochastic process $\{S_n : n = 0, 1, 2, \dots\}$ where $S_n = \sum_{i=1}^n X_i$ and $X_1, X_2, \dots$ are iid with $P(X_i = 1) = P(X_i = -1) = 0.5$. This is called a symmetric random walk since the probability of moving up and down in each step is the same.

Random walks can be generalized to continuous time, non symmetric walks, several dimensions, etc.

A random walk process is very easy to simulate, see the book.